

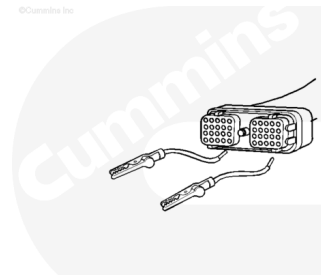
Trainings ▾

Resistance Check

⚠ CAUTION ⚠

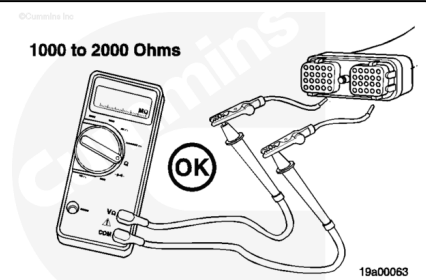
Do not use probes or leads other than Part No. 3822758. The connector will be damaged. The leads must fit tight in the connector without expanding the pins of the connector.

Insert the one of the leads into pin 21 of the engine harness adaptor cable connector. Insert the other lead into pin 22.



Make sure the ESS is connected to the sensor harness.

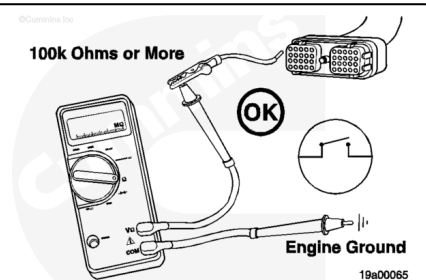
Measure the resistance. The resistance value **must** be less than 1500 ohms. If the resistance is **not** correct, there is a problem with the engine harness, provided the sensor was previously checked. Refer to Procedures 019-202 (</qs3/pubsys2/xml/en/procedures/94/94-019-202.html>), 019-240 (</qs3/pubsys2/xml/en/procedures/94/94-019-240.html>), and 019-043 (</qs3/pubsys2/xml/en/procedures/94/94-019-043.html>).



Check for Short Circuit to Ground

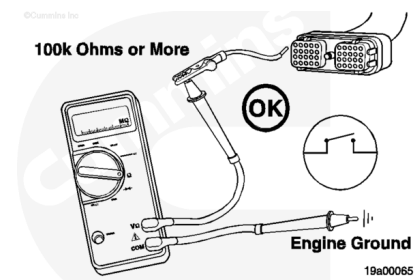
Disconnect the engine harness adaptor cable from the ECM.

Insert the lead into pin 21. Touch the other Multimeter probe to the engine block. Measure the resistance. The Multimeter **must** show an open circuit (more than 100k ohms).



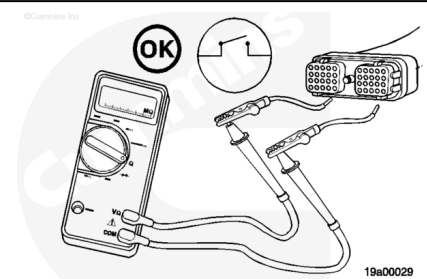
Remove the lead from pin 21 and place it into pin 22. Touch the other Multimeter probe to the engine block. Measure the resistance. The Multimeter **must** show an open circuit (more than 100k ohms).

If the resistance values in any of the previous checks are **not** within the specification, there is a short circuit to ground, provided the ESS sensor has been previously checked. Repair or replace the engine harness. Refer to Procedure 019-199 (</qs3/pubsys2/xml/en/procedures/94/94-019-199.html>), 019-240 (</qs3/pubsys2/xml/en/procedures/94/94-019-240.html>) or 019-043 (</qs3/pubsys2/xml/en/procedures/94/94-019-043.html>).



Check for Short Circuit from Pin to Pin

Disconnect the engine harness adaptor cable from the ECM. Insert one of the multimeter leads into pin 21. Check the resistance to all pins in the connector. The multimeter **must** show an open circuit (more than 100k ohms).



Remove the lead from pin 21 and place it into pin 22, and again check the resistance to all pins in the connector. The multimeter **must** show an open circuit (more than 100k ohms) at all pins.

If the resistance values in any of the previous checks are **not** within specification, there is a short circuit from pins 21 or 22 to any pin that measures less than 100k ohms. Repair or replace the engine harness. Refer to Procedures 019-199 (</qs3/pubsys2/xml/en/procedures/94/94-019-199.html>), 019-240 (</qs3/pubsys2/xml/en/procedures/94/94-019-240.html>), or 019-043 (</qs3/pubsys2/xml/en/procedures/94/94-019-043.html>).

